

Portfolio Case Studies

Voice Card

Clear, honest, practical, always learning

Before vs After

Generic AI Version

I developed an innovative web application using modern technologies to provide users with a seamless and engaging experience.

My Edited Version

I built BookNest to turn my frontend learning into a practical application. It helped me understand how real projects work — from connecting APIs and handling data to improving user experience through responsive design.

Case Study: BookNest – Book Finder & Reading Tracker

Problem

While learning frontend development, I wanted to move beyond tutorials and apply my knowledge by building a real application. I wanted a project that would challenge me to work with dynamic data, improve my JavaScript skills, and understand how different parts of a web application connect together.

BookNest was created during my Zyora Labs Frontend Internship as an opportunity to practice building a complete frontend project with real-world features instead of only creating static pages.

What I Built & Decisions I Made

BookNest is a responsive web application that helps users discover books, explore book information, and manage their reading journey.

I built the application using **HTML, CSS, and JavaScript** and focused on creating a smooth experience for users.

Key features and decisions:

- Integrated the **Google Books API** to allow users to search through a large collection of books.
- Added the **Open Library API as a fallback option** to improve reliability when the primary API does not return expected results.
- Designed responsive layouts so the application works across different screen sizes.
- Created book search functionality with loading states, validation, and error handling.
- Added reading tracker features to help users manage books they want to read and books they have completed.
- Improved the project through continuous testing, debugging, documentation, and deployment.

One of my biggest learning challenges was JavaScript because I was completely new to it when I started. While building BookNest, I had to understand how JavaScript connects with HTML, manages user actions, and handles data from APIs.

A major technical challenge was integrating the Google Books API. Initially, the API connection was not working properly, and the expected book data was not displaying. I debugged the issue, learned more about API requests, tested different approaches, and eventually got the data flow working correctly.

One decision I am especially proud of was adding an API fallback. Instead of depending on only one data source, I considered reliability and created a better experience for users.

Outcome & What I Learned

BookNest became more than just an internship assignment — it became a project where I learned how frontend applications work in practice.

Through this project, I improved my understanding of:

- JavaScript fundamentals
- API integration
- Data handling
- Responsive web design

- Debugging and problem-solving
- Git and project documentation

The biggest takeaway from this project was learning by building. Since many concepts were new to me, every challenge became an opportunity to improve. The experience and skills I gained from BookNest will help me approach future projects with more confidence.

About Me

Hi, I'm **Thillai Eswari**, a Computer Science Engineering student passionate about building practical software projects and continuously improving my development skills.

I enjoy learning through hands-on experience and exploring areas like backend development, AI, and web technologies. I believe every project is an opportunity to learn something new, solve problems, and become a better developer.

Let's Connect

I enjoy connecting with people who are interested in technology, learning, and building meaningful projects.

Feel free to explore my work, share feedback, or connect with me through GitHub and LinkedIn.

Linkedin: <https://linkedin.com/in/thillaieswarit26/>

Github: <https://github.com/thillaieswari26/>